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Project Title:	Understanding Full-Body Gaze Control During Locomotion		
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Cohen, Helen (Baylor College of Medicine) Mulavara, Ajitkumar (NSBRI)		
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Control of locomotion requires precise interaction between several sensorimotor subsystems. Exposure to the microgravity environment of space flight leads to postflight adaptive alterations in these multiple subsystems leading to postural and gait disturbances along with decrement in visual acuity during body motion. Countermeasures designed to mitigate these postflight gait alterations need to be assessed with a new generation of functional tests that evaluate the interaction of various elements central to locomotor control. Traditionally, gaze stabilization has been studied almost exclusively as a problem of eye-head-trunk coordination. However, coordination between the eye-head and trunk are not the only mechanism aiding gaze stabilization particularly during locomotion. Therefore the first goal of this study was to develop new techniques to evaluate the multiple, interdependent, full- body sensorimotor subsystems that aid gaze stabilization during locomotion. The second goal was to develop new tests to evaluate changes in dynamic visual acuity

Task Description:

during body movement. To address the first goal, two experiments were performed. In the first study we investigated how alteration in gaze tasking changes full-body locomotor control strategies. Subjects performed two discreet gaze stabilization tasks while walking on a motorized treadmill: 1) focusing on a central point target; 2) reading numeral characters; both presented at 2m in front at eye level. The second study investigated the potential of adaptive remodeling of the full-body gaze control systems following exposure to visual-vestibular conflict known to adaptively modify vestibulo-ocular reflex (VOR) function. Subjects walked on the treadmill before and after they were exposed to 0.5X minifying lenses worn for 30 minutes during self-generated sinusoidal vertical head rotations performed while seated. In both studies we measured: temporal parameters of gait, full body sagittal plane segmental kinematics of the head, trunk, thigh, shank and foot, accelerations along the vertical axis at the head and the shank, and the vertical forces acting on the support surface. Results showed that while reading numeral characters as compared to the central point target: 1) compensatory head pitch movements increased, 2) the peak acceleration measured at the head was significantly reduced, 3) the knee joint total movement was on greater during the period from the heel strike event to the peak knee flexion event in stance phase of the gait cycle. Results also indicate that following exposure to visual-vestibular conflict known to adaptively modify the VOR subjects predominantly: 1) decreased the amplitude of head movement with respect to space; 2) increased the amount of knee and ankle flexion during the initial stance phase of the gait cycle. These responses serve to aid gaze stabilization during locomotion. Taken together, results from Studies 1 and 2 provide evidence that the full body contributes to gaze stabilization during locomotion, and that different functional elements are responsive to changes in visual task constraints and are subject to adaptive alterations following exposure to visual-vestibular conflict. These studies successfully validated new integrated methodologies designed to assess locomotor function for countermeasure evaluation and validation. To address the second goal of this study we developed a new test to measure dynamic visual acuity during treadmill walking. Upon their return to earth, astronauts experience the effects of vestibular adaptation to microgravity. The postflight changes in vestibular information processing affects postural and locomotor stability and leads to reduced visual acuity during body movement due to alterations in gaze control. These changes in acuity have significant operational implications. The inability to see clearly during body motion can impair the ability to operate spacecraft, conduct EVAs and perform an emergency egress soon after landing following a long-duration space flight. However, it is likely that time spent in microgravity affects canal and otolith function differently. As a result, the isolated rotational stimuli used in traditional tests of canal function may fail to identify vestibular deficits after space flight. In a gaze control task, the relative contributions of the canal and otolith organs are modulated with viewing distance. The ability to stabilize gaze during a perturbation, on visual targets placed at different distances from the head may therefore provide independent insight into the function of these systems. Our newly developed dynamic visual acuity test allows us to measure changes both in static and dynamic visual acuity for both near (0.5 m) and far (4 m) visual target positions. This test was evaluated in both normal subjects and in patients with bilateral vestibular impairment. Results show a significant ability to reliably differentiate normal from clinical behavior. We have used results obtained from this research to develop an inflight measure of dynamic visual acuity. This test will measure static visual acuity while subjects stand on the ISS treadmill and dynamic visual acuity during treadmill walking. We are currently developing an integrated testing system using a computer driven microdisplay screen. This inflight test will measures changes in static and dynamic visual acuity during the initial adaptation phase to spaceflight and during the full duration of the flight. This newly developed test will be performed on the ISS following delivery of the hardware via an upcoming Shuttle flight. Both our full-body treadmill locomotion and near/far dynamic visual acuity tests have been accepted by the Clinical Status Evaluation (CSE) review committee and are now going through the final approval process to include these evaluation techniques as part of the standard complement of tests that all astronauts undergo to evaluate their recovery.
Traditionally, gaze stabilization has been studied almost exclusively as a problem of eye-head-trunk coordination. However, coordination between the eye-head and trunk may not be the only mechanism aiding gaze stabilization particularly during locomotion. Therefore the first goal of this study was to determine how the multiple, interdependent, full- body sensorimotor subsystems aiding gaze stabilization during locomotion are functionally coordinated. The second goal was to use this information to develop new tests of locomotion function to be used to evaluate the efficacy of countermeasures.

To address the first goal, two experiments were performed. In the first study (Study 1) we investigated how alteration in gaze tasking changes full-body locomotor control strategies. Subjects (n=9) performed two discreet gaze stabilization tasks while walking at 6.4 km/hr on a motorized treadmill: 1) focusing on a central point target; 2) reading numeral characters; both presented at 2m in front at eye level. The second study (Study 2) investigated the potential of adaptive remodeling of the full-body gaze control systems following exposure to visual-vestibular conflict. Subjects (n=14) walked (6.4 km/h) on the treadmill before and after they were exposed to 0.5X minifying lenses worn for 30 minutes during self-generated sinusoidal vertical head rotations performed while seated. In both studies we measured: temporal parameters of gait, full body sagittal plane segmental kinematics of the head, trunk, thigh, shank and foot, accelerations along the vertical axis at the head and the shank, and the vertical forces acting on the support surface.

Results from Study 1 showed that while reading numeral characters as compared to the central point target: 1) compensatory head pitch movements increased, 2) the peak acceleration measured at the head was significantly reduced, 3) the knee joint total movement was on greater during the period from the heel strike event to the peak knee flexion event in stance phase of the gait cycle. Results from Study 2 indicate that following exposure to visual-vestibular conflict changes in full-body strategies were observed consistent with the requirement to aid gaze stabilization during locomotion.

Taken together, results from Studies 1 and 2 provide evidence that the full body contributes to gaze stabilization during locomotion, and that different functional elements are responsive to changes in visual task constraints and are subject to adaptive alterations following exposure to visual-vestibular conflict. These studies also successfully validate new integrated methodologies designed to assess locomotor function for countermeasure evaluation and validation.

To address the second goal of this study we developed a new test to measure dynamic visual acuity during treadmill walking. Astronauts returning from spaceflight experience reduced visual acuity during body motion of the kind experienced during walking due to alterations in gaze stability caused by neurovestibular adaptive changes. These changes in acuity have significant operational implications. The inability to see clearly during body motion can impair the ability to operate spacecraft, conduct EVAs and perform an emergency egress soon after landing following a long-duration spaceflight. Our newly developed dynamic visual acuity test allows us to measure changes both in static and dynamic visual acuity for both near (0.5 m) and far (4 m) visual target positions. This test was evaluated in both normal subjects and in patients with bilateral vestibular impairment. Results show a significant ability to reliably differentiate normal from clinical behavior.

	We have used results obtained from this research to develop an inflight measure of dynamic visual acuity. This test will measure static visual acuity while subjects stand on the ISS treadmill and dynamic visual acuity during treadmill walking. We are currently developing an integrated testing system using a computer driven microdisplay screen. This inflight test will measures changes in static and dynamic visual acuity during the initial adaptation phase to spaceflight and during the full duration of the flight. This newly developed test will be performed on the ISS following delivery of the hardware via Shuttle flight ULF-2 in 2005.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our experiments are yielding results that may be compared and contrasted with the impairment experienced by the elderly or clinical populations. The investigation of neural adaptation to microgravity will lead to better understanding of neural alterations associated with aging and other neurological disorders. The development of unique research protocols to investigate neural alterations in the control of gaze can aid clinicians in diagnosis of neurological and neurovestibular pathology and in monitoring post-surgical recovery. The development of a unique research protocol to determine how normal subjects adapt to altered sensory information can be used by clinicians to develop enhanced rehabilitation techniques for patients with balance disorders. Development of this new technology can lead to the establishment of worldwide clinical vestibular testing norms that can be used in medical facilities. In addition, this research can lead to the formulation of models of neural activity based on known pathways and substrates. These models can be used to make predictions about response properties of a variety of motor subsystems following exposure to microgravity or as a predictive tool for clinical populations. Specifically, our newly developed integrated tests of locomotor function serve to better assess how multiple sensorimotor systems change response characteristics both in aging and during various disease states. Our NEAR/FAR dynamic visual acuity test serves as a very sensitive marker of vestibular dysfunction. Decrement in dynamic visual acuity is a common complaint of vestibular deficient patients and through this project we have developed a unique method to characterize these changes but also to use these data as a tool to guide rehabilitation interventions. The development of a unique research protocol to determine how normal subjects adapt to altered sensory information can be used by clinicians to develop enhanced rehabilitation techniques for patients with balance disorders. Development of this new technology can lead to the establishment of worldwide clinical vestibular testing norms that can be used in medical facilities. In addition, this research can lead to the formulation of models of neural activity based on known pathways and substrates. These models can be used to make predictions about response properties of a variety of motor subsystems following exposure to microgravity or as a predictive tool for clinical populations.
Task Progress:	To address the second goal of this study we developed a new test to measure dynamic visual acuity during treadmill walking. Upon their return to earth, astronauts experience the effects of vestibular adaptation to microgravity. The postflight changes in vestibular information processing affects postural and locomotor stability and leads to reduced visual acuity during body movement due to alterations in gaze control. These changes in acuity have significant operational implications. The inability to see clearly during body motion can impair the ability to operate spacecraft, conduct EVAs and perform an emergency egress soon after landing following a long-duration space flight. However, it is likely that time spent in microgravity affects canal and otolith function differently. As a result, the isolated rotational stimuli used in traditional tests of canal function may fail to identify vestibular deficits after space flight. Also, the functional and operational consequences of deficits that are identified often remain unknown. In a gaze control task, the relative contributions of the canal and otolith organs are modulated with viewing distance. The ability to stabilize gaze during a perturbation, on visual targets placed at different distances from the head may therefore provide independent insight into the function of these systems. Our newly developed dynamic visual acuity test allows us to measure changes both in static and dynamic visual acuity for both near (0.5 m) and far (4 m) visual target positions. This test was evaluated in both normal subjects and in patients with bilateral vestibular impairment. Results show a significant ability to reliably differentiate normal from clinical behavior. We have used results obtained from this research to develop an inflight measure of dynamic visual acuity. This test will measure static visual acuity while subjects stand on the ISS treadmill and dynamic visual acuity during treadmill walking. We are currently developing an integrated testing system using a computer driven microdisplay screen. This inflight test will measures changes in static and dynamic visual acuity during the initial adaptation phase to spaceflight and during the full duration of the flight. This newly developed test will be performed on the ISS following delivery of the hardware via an upcoming Shuttle flight.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Mulavara AP, Houser J, Miller C, Bloomberg JJ. "Full-body gaze control mechanisms elicited during locomotion: effects of VOR adaptation." J Vestib Res. 2005;15(5-6):279-89. PMID: 16614474 , Jun-2005
Articles in Peer-reviewed Journals	Peters BT, Bloomberg JJ. "Dynamic visual acuity using FAR and NEAR targets." Acta Otolaryngol. 2005 Apr;125(4):353-7. PMID: 15823804 , Apr-2005
Awards	Bloomberg J. "NSBRI Associate Team Lead: Sensorimotor, JJ Bloomberg, January 2004." Jan-2004
Presentation	Bloomberg J. J., Mulavara A. P., Cohen H. S., Richards J. T., Miller, C. A. Peters, B. T., Marshburn A. M., Brady "Developing tests to evaluate astronaut postflight dysfunction" N/A Jan-2004
Presentation	Bloomberg J. J., Mulavara A. P., Peters B. T., Cohen H. S., Landsness E. C., Black, F. O "Integrated locomotor function tests for countermeasure evaluation" N/A Jan-2005
Presentation	Bloomberg JJ, Mulavara AP "Understanding astronaut locomotor dysfunction after spaceflight: a sensorimotor perspective" N/A Sep-2004

Presentation	Bloomberg JJ, Mulavara AP, Cohen HS "Functional coordination of a full-body gaze control mechanisms elicited during locomotion." N/A Jan-2004
Presentation	Bloomberg JJ, Mulavara AP, Kozlovskaya "Adaptive reorganization of locomotor function after long-duration spaceflight" N/A Jun-2004
Presentation	Bloomberg JJ, Mulavara AP, Kozlovskaya IB "Reorganization of locomotor strategies after long-duration spaceflight" N/A May-2004
Presentation	Landsness E. C., Bloomberg J. J., Peters B. T., Black F. O "Far and near target dynamic visual acuity: bilateral vestibular deficit patients" N/A Jul-2004
Presentation	Landsness E. C., Bloomberg J. J., Peters B. T., Black F. O. "FAR and NEAR Target Dynamic Visual Acuity: Bilateral Vestibular Deficit Patients" N/A Feb-2005
Presentation	Miller CA, Mulavara AP, Richards JT, Peters BT, Houser J, Marshburn AM, Bloomberg JJ "Coordination of lower-limb joints during locomotion: the effects of vestibulo-ocular reflex adaptation" N/A Oct-2004
Presentation	Mulavara A. P., Houser J., Peters B. T., Miller C. A., Richards, J. T., Marshburn A. Brady R., Cohen H., Bloomberg J. J "Full-body gaze control mechanisms elicited during locomotion: effects of VOR adaptation" N/A Jul-2004
Presentation	Mulavara AP, Bloomberg JJ "Identifying head-trunk and lower limb contributions to gaze stabilization during locomotion" N/A Oct-2004
Presentation	Mulavara AP, Miller CA, Houser J, Richard JT, Bloomberg JJ "Gaze stabilization during locomotion requires full body coordination" N/A Aug-2004
Presentation	Peters B. T., Brady R. A., Landsness, E. C., Black F. O., Bloomberg J. J "Far and near target dynamic visual acuity: a functional assessment of canal and otolith performance" N/A Jul-2004
Presentation	Peters BT, Bloomberg, JJ "Dynamic visual acuity during locomotion using far and near targets" N/A Oct-2004